

UrbanTwin: Market-Sim

Generative Agent-Based Spatial Planning & Multi-Stakeholder Consensus Engine

EVENT	LOCATION	BUILD WINDOW	TEAM COMPOSITION
Claude Impact Lab #2	Fraunhofer IIS, Nürnberg	7 Hours (13:00–20:00)	4 Engineers / Designers

Part 1: The Layman's Explanation (The Story)

The Elevator Pitch: "Nuremberg's vegetable market needs a new home, but every human stakeholder is fighting over where to put it. Instead of spending 2 years on expensive surveys, UrbanTwin uses AI to simulate a living digital Nuremberg. In 10 seconds, hundreds of AI citizens walk the streets, shop, and deliver goods, showing city planners exactly which location creates harmony—and where the bottlenecks will happen."

1. What is the real-world problem?

Nuremberg's famous vegetable market sits on the central square (*Hauptmarkt*). Due to urban events and logistics, the city must relocate it. Two top candidates are being debated:

- **Lorenzkirche:** A bustling pedestrian plaza outside the historic St. Lorenz church, directly at a major subway stop.
- **Former Kaufhof Building:** The empty ground floor of a defunct department store in downtown Nuremberg.

The catch: No single location makes everyone happy. A place with huge pedestrian foot-traffic might be impossible for vegetable delivery trucks to reach at 5:00 AM. A place with great indoor shelter might feel too far to walk for seniors.

2. Meet the AI Citizens (Simulated Personas)

UrbanTwin doesn't output dry statistics. It simulates real demographic archetypes with goals, budgets, and physical limitations:

 Oma Helga (Age 78) · Mobility Constrained Takes the U-Bahn, uses a rolling cart. Highly sensitive to stairs, cobblestones, and walking distances over 250m. Loves Kaufhof (flat indoor ground, elevators); rejects Lorenzkirche if crowds block her way.	 Markus (Age 45) · Market Produce Vendor Drives a 3.5-ton Sprinter van at 5:30 AM with 40 crates of regional vegetables. Needs loading dock clearance, turning radius, and zero bollard friction. Kaufhof has loading docks; Lorenzkirche has delivery nightmares.
 Lukas (Age 29) · Fast-Paced Commuter Has a 30-minute lunch break. Exits U-Bahn Lorenzkirche. Prioritizes speed and proximity. Loves Lorenzkirche because it's right on his commute route; won't walk 10 minutes out of his way to Kaufhof.	 Frau Weber · Neighboring Boutique Retailer Owns a clothing store nearby. Worried about smelly vegetable waste and blocked storefront windows, but welcomes high footfall that brings secondary sales into her boutique.

3. What Does the Finished Tool Do?

A city planner opens UrbanTwin on a browser, drags the market footprint onto Lorenzkirche or Kaufhof, and presses **"Simulate Saturday Peak"**. Instantly, visual agents navigate the map. A live **Consensus Index (0–100%)** calculates satisfaction across all 4 groups, while a spatial heatmap glows red where pedestrian or vehicular congestion occurs.

Part 2: Deep Technical Architecture & System Design

To satisfy Fraunhofer IIS researchers and Anthropic judges, this project must not be a superficial chatbot. It is built as a **Coupled Agent-Based Spatial Graph Engine**.

Layer	Technology	Architectural Responsibility
Spatial Graph	Leaflet / Mapbox GL + OpenStreetMap (OSM)	Nodes & edges representing pedestrian walkways, U-Bahn elevator coordinates, delivery vehicle restricted zones in central Nuremberg.
Pathfinding & Isochrones	Turf.js + Dijkstra Graph Search	Calculates exact walking distance, elevation friction, and 3-minute / 5-minute walking time isochrone envelopes for both sites.
Simulation Engine	Monte Carlo Agent State Machine (TypeScript)	Runs 100 stochastic agents through a day cycle (arrival, path selection, queuing, purchase, exit). Records friction points & drop-offs.
Claude Cognitive Brain	Anthropic Claude 3.5 Sonnet (Tool Use)	Ingests aggregated simulation metrics and acts as the municipal consensus solver: resolves trade-offs, evaluates retail synergies, and drafts council policy briefs.

System Data Flow & Communication Contract

The backend produces a standardized `SimulationResult` JSON consumed by the dashboard:

```
{
  "location": "Kaufhof_GroundFloor",
  "consensusScore": 79.4,
  "stakeholderScores": { "seniors": 92.0, "commuters": 64.5, "vendors": 88.0, "retailers": 73.0 },
  "criticalBottlenecks": [{ "zone": "South_Portal_Elevator", "congestionProbability": 0.84, "reason": "Rollator queue exceeds capacity at 11:30" }],
  "claudeRecommendation": "Optimal site for logistics and demographic inclusivity. Mitigate commuter drop-off by deploying 3 rapid-service fruit kiosks outside entrance."
}
```

Part 3: 7-Hour Execution Sprint (The 4-Person Playbook)

Time Window	Phase Objective	Team Coordination & Deliverables
13:00 – 13:30 (30 mins)	Architecture Freeze & Contract Setup	- Lock the JSON schema between UI and simulation engine. - Set up GitHub repo, Vite + React scaffolding, install <code>@anthropic-ai/sdk</code>, Leaflet/Mapbox. - Download GeoJSON boundaries for Nuremberg Altstadt, Lorenzkirche, and Kaufhof.
13:30 – 16:30 (3.0 hours)	Decoupled Deep Build (Silent Sprint)	Person 1 (Spatial UI): Map view of Nuremberg with toggle between Hauptmarkt, Lorenzkirche, and Kaufhof. Animated agent dots. Person 2 (Analytics UI): Scorecard dials (Senior Accessibility, Logistics Clearance, Retail Synergy) + live citizen quote stream. Person 3 (Sim Engine): Generates 100 agent schedules, runs pathfinding algorithms, computes travel friction metrics. Person 4 (Claude LLM): Constructs Claude prompt with Tool Calling; transforms sim statistics into executive council decisions.
16:30 – 17:00	Dinner Break 🍽️	Align integration endpoints; identify any remaining bugs.
17:00 – 18:00 (1.0 hour)	System Integration & Polishing	- Connect Frontend to Sim Engine and Claude API. - Stress-test the demo: verify clicking "Run Simulation" returns in < 4 seconds. - Style with sleek dark-mode civic aesthetic (badges, glowing conflict zones).
18:00 – 18:30 (30 mins)	Pitch Rehearsal & Deck Freeze	- Rehearse the strict 2-minute pitch 4 times with stopwatch. - Prepare a backup screen recording in case venue Wi-Fi stutters.

Part 4: Individual Role Breakdown & Deliverables

Person 1 Spatial Map & Visual Simulation Lead (Frontend 1)

Primary Tooling: React / Next.js, Leaflet / Mapbox GL JS, TailwindCSS.

- Renders high-contrast map of central Nuremberg showing the 3 contender locations.
- Animates 30–50 moving SVG dots (blue = commuters, purple = seniors, orange = delivery vans) along real road/walkway coordinates.
- Displays glowing red warning polygons over spatial bottleneck zones.

Person 2 Stakeholder Analytics & Dashboard Lead (Frontend 2)

Primary Tooling: Recharts, Lucide Icons, Framer Motion.

- Builds the "Consensus Cockpit": radial progress gauges for each of the 4 demographic archetypes.
- Creates the "Citizen Voice" feed displaying dynamically generated live quotes (e.g., *Oma Helga: "Elevator at Kaufhof is wonderful!"*).
- Adds an "Export Municipal Brief" button triggering an automated PDF/Markdown report.

Person 3 Spatial Graph & Simulation Logic Lead (Backend / Data)

Primary Tooling: TypeScript / Python, Turf.js, GeoJSON.

- Prepares exact geographic distance matrices: U-Bahn stops to Lorenzkirche vs. Kaufhof.
- Implements the agent friction model: calculates drop-off rates when walking distances exceed thresholds (e.g., 250m for elderly).
- Packages the quantitative metrics and passes them via JSON to the Claude reasoning service.

Person 4 AI Architect & Claude Reasoning Lead (AI / Pitch Lead)

Primary Tooling: Anthropic Claude SDK (Python or Node), Structured JSON Outputs.

- Engineers system prompts for Claude 3.5 Sonnet to evaluate spatial trade-offs objectively.
- Builds structured tool calls: `evaluate_tradeoffs()`, `generate_mitigation_plan()`.
- Owns the 2-minute pitch deck, script narrative, and timing.

Part 5: The Winning 2-Minute Pitch Script

[0:00 - 0:25] The Hook:

"Judges, Nuremberg's vegetable market has to move from the Hauptmarkt. But if you put it at Lorenzkirche, delivery trucks can't get in at 5:00 AM. If you put it in the former Kaufhof, office commuters complain it's too far for a quick lunch. For two years, city halls debate with surveys that take months and please nobody."

[0:25 - 0:50] The Solution:

"Meet UrbanTwin: Market-Sim. We don't guess. We built an AI-powered digital twin of Nuremberg. Using Claude 3.5 Sonnet and spatial graph models, we simulate hundreds of digital citizens—from 78-year-old Oma Helga with her rolling cart, to delivery drivers in 3.5-ton Sprinters—living through a market Saturday before a single stall is built."

[0:50 - 1:35] The Live Demonstration:

(Presenter clicks on Lorenzkirche)

"Look at Lorenzkirche: Commuter happiness is 94%, but watch the vendor score plummet to 28%—the delivery vans are completely bottlenecked by pedestrian bollards."

(Presenter clicks on Kaufhof)

"Now we switch to Kaufhof: Vendor score skyrockets to 92% thanks to loading docks. Oma Helga's accessibility is 95%. But Claude spots a critical flaw: lunch commuters drop by 35%. In 2 seconds, Claude synthesizes a compromise: add 3 express fruit stalls facing the street, and consensus hits 88% across all groups."

[1:35 - 2:00] The Close:

"This isn't a chatbot. It's an evidence-based consensus engine that turns subjective political fights into objective, data-backed city planning. Built with Claude, ready for Nuremberg. Thank you."